Performance Metrics

Classification Performance Metrics

Used when output is categorical.

Examples:

- Spam Detection
- Disease Prediction
- Sentiment Analysis
- Fraud Detection

Evaluation is based on a **Confusion Matrix**.

- A confusion matrix is used to evaluate the performance of a classification model.
- It compares:
  - Actual class (Target)
  - Predicted class (Output)
- Rows represent the actual classes.
- Columns represent the predicted classes.
- Correct classifications appear on the main diagonal.
- Misclassifications appear outside the diagonal.

Suppose we have two classes: $(\mathrm{C}_1,\ C_2)$

	Predicted Class	
Actual Class	C_1	C_2
C_1	5	1
C_2	1	4

Confusion Matrix Terminology

Actual	Predicted	
	Positive (1)	Negative (0)
Positive (1)	TP	FN
Negative (0)	FP	TN

- **TP (True Positive):** Actual $= 1$, Predicted $= 1$
- **TN (True Negative):** Actual $= 0$, Predicted $= 0$
- **FP (False Positive):** Actual $= 0$, Predicted $= 1$
- **FN (False Negative):** Actual $= 1$, Predicted $= 0$

$$TP + TN = \text{Correct Predictions}$$

$$FP + FN = \text{Misclassifications}$$

Accuracy

Accuracy measures the fraction of correctly classified samples.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Class Imbalance

A dataset is said to be **imbalanced** when the number of instances belonging to one class is significantly larger than the other class(es).

Example: Spam Detection Dataset

Class	Number of Emails
Not Spam	900
Spam	100
Total	1000

Class Ratio:

$$900 : 100 = 9 : 1$$

Since one class dominates the other, this is an **Class Imbalanced Dataset**.

Why Accuracy Can Be Misleading?

Suppose a model predicts every email as "Not Spam"

Actual Class	Predicted Class
Not Spam(900)	Not Spam
Spam(100)	Not Spam

Correct Predictions: 900
Total Predictions:1000
Accuracy:

$$Accuracy = \frac{900}{1000} = 0.9 = 90\%$$

Problem:

- Accuracy is very high (90%).
- The model failed to detect even a single spam email.
- Therefore, accuracy alone is not suitable for imbalanced datasets.
- Metrics such as Precision, Recall, F1-score and MCC are preferred.

Performance Measure: Precision

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Definition

- Measures how many of the predicted positive instances are actually positive.
- High precision is important when false positives (false alarms) are costly.

Example : Spam Email Detection

- If an email is predicted as spam, it should truly be spam.
- Incorrectly marking an important email as spam (false positive) should be minimized.

Performance Measure: Recall

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Definition

- Measures how many of the actual positive instances are correctly identified.
- Out of all actual positive instances, how many are correctly detected?
- High recall is important when false negatives (missed positives) are dangerous.

Example : Disease Detection

- The model should identify as many patients with the disease as possible.
- Missing a patient who has the disease (false negative) can have serious consequences.

Precision and Recall Example

	Actual Positive	Actual Negative
Predicted Positive	40	5
Predicted Negative	10	45

$$TP = 40, \quad FP = 10, \quad FN = 5, \quad TN = 45$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{40 + 45}{100} = 85\%$$

$$Precision = \frac{TP}{TP + FP} = \frac{40}{50} = 80\%$$

$$Recall = \frac{TP}{TP + FN} = \frac{40}{45} = 88.9\%$$

Multiclass Accuracy

$$\text{Accuracy} = \frac{TP_A + TP_B + TP_C}{\sum_i \sum_j m_{ij}}$$

or equivalently,

$$\text{MCC} = \frac{TP_A + TP_B + TP_C}{N}$$

where

$$N = \sum_i \sum_j m_{ij}$$

For the confusion matrix:

$$\text{Correct Predictions}(Diagonalelements) = 5 + 4 + 4 = 13$$

Suppose we have three classes: (C_1, C_2, C_3)

Actual Class	Predicted Class C_1	C_2	C_3
C_1	5	1	0
C_2	1	4	1
C_3	2	0	4

$$\text{Total Predictions} = 5 + 1 + 0 + 1 + 4 + 1 + 2 + 0 + 4 = 18$$

$$\text{Accuracy} = \frac{13}{18} = 0.722$$

$$\boxed{\text{Accuracy} = 72.2\%}$$

Precision Calculation

Class A :

$$Precision_A = \frac{5}{5+1+2} = \frac{5}{8} = 0.625$$

Class B

$$Precision_B = \frac{4}{1+4+0} = \frac{4}{5} = 0.80$$

Class C

$$Precision_C = \frac{4}{0+1+4} = \frac{4}{5} = 0.80$$

$$Precision_{macro} = \frac{0.625+0.80+0.80}{3} = 0.742$$

$\boxed{\text{Macro Precision} = 74.2\%}$

Recall Calculation

Class A

$$Recall_A = \frac{5}{5+1+0} = \frac{5}{6} = 0.833$$

Class B

$$Recall_B = \frac{4}{1+4+1} = \frac{4}{6} = 0.667$$

Class C

$$Recall_C = \frac{4}{2+0+4} = \frac{4}{6} = 0.667$$

$$Recall_{macro} = \frac{0.833 + 0.667 + 0.667}{3} = 0.722$$

$$\boxed{\text{Macro Recall} = 72.2\%}$$

F1 Score

Combines Precision and Recall.

$$F_1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right)$$

Interpretation

- High only if both precision and recall are high.
- Useful for imbalanced datasets.

F-Beta Score

General form:

$$F_\beta = (1 + \beta^2)\frac{PR}{\beta^2 P + R}$$

- $\beta = 1$: F1 Score
- $\beta < 1(0.5)$: Precision prioritized
- $\beta > 1(2)$: Recall prioritized

Interpretation

- If False positive and False negative both are equally important

Matthews Correlation Coefficient (MCC)

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Range

$$-1 \leq MCC \leq 1$$

- 1 = Perfect prediction
- 0 = Random prediction
- -1 = Total disagreement

Interpretation

- takes into account TP, TN, FP and FNs and is generally regarded as a balanced measure even when the classes are imbalanced

ROC Curve

ROC:

- Receiver Operating Characteristic Curve helps in visualizing the performance of a model.
- It shows the efficiency of a model in the detection of TPs while avoiding the occurrence of FPs.

TPR (Type I error): Measures how many positives are correctly found

$$TPR = \frac{TP}{TP + FN}$$

FPR (Typr II error): Measures how many negatives are incorrectly labeled as positive.

$$FPR = \frac{FP}{FP + TN}$$

Plot:TPR vs FPR

Ideal classifier: (0,1)

The ideal point on the ROC curve is (0, 1), representing 0% false positives and 100% true positives.

ROC Curve: Basic Idea

A binary classifier predicts a **probability** for each instance rather than directly predicting the class. **Example Dataset**

Actual Class (y)	Predicted Probability ($\hat{y}$)
1	0.8
0	0.96
1	0.4
1	0.3
0	0.2
1	0.7

The classifier outputs probabilities. A **threshold** is used to convert these probabilities into class labels.

$$\hat{y} = \begin{cases} 1, & \hat{y} \geq \text{Threshold} \\ 0, & \hat{y} < \text{Threshold} \end{cases}$$

Step 1: Choose Threshold

Suppose the threshold values are

$$[0,\ 0.2,\ 0.4,\ 0.6,\ 0.8]$$

For each threshold,

1. Convert probabilities into class labels.
2. Construct the confusion matrix.
3. Compute

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

4. Plot the point (FPR, TPR).

Repeating this process for all thresholds produces the ROC curve.

Threshold $= 0$

Decision Rule

$$\hat{y} \geq 0$$

Since every probability is greater than or equal to zero,

$$\hat{y} = [1, 1, 1, 1, 1, 1]$$

Confusion Matrix:TP=4, FN=0
FP=2, TN=0 Therefore,
TPR=

$$\frac{4}{4} = 1$$

$$FPR = \frac{2}{2} = 1$$

ROC Point :(1,1)

Threshold $= 0.2$

Decision Rule

$$\hat{y} \geq 0.2$$

Predicted labels

$$[1, 1, 1, 1, 1, 1]$$

Confusion Matrix

$$TP = 4, \quad FN = 0$$

$$FP = 2, \quad TN = 0$$

Hence,

$$TPR = 1, \qquad FPR = 1, \qquad ROCpoint = (1, 1)$$

Note: If the rule is $\hat{y} > 0.2$, then the instance with probability 0.2 is classified as Negative, giving

$$FP = 1, \qquad TN = 1$$

and

$$FPR = \frac{1}{2} = 0.5.$$

Threshold $= 0.4$

Predicted labels : [1,1,1,0,0,1]
Confusion Matrix

$$TP = 3, \quad FN = 1$$

$$FP = 1, \quad TN = 1$$

Therefore,

$$TPR = \frac{3}{4} = 0.75$$

$$FPR = \frac{1}{2} = 0.5$$

ROC Point

$$(0.5,\ 0.75)$$

Threshold $= 0.6$

Predicted labels [1,1,0,0,0,1]
Confusion Matrix

$$TP = 2, \quad FN = 2$$

$$FP = 1, \quad TN = 1$$

Therefore,

$$TPR = \frac{2}{4} = 0.5$$

$$FPR = \frac{1}{2} = 0.5$$

ROC Point

$$(0.5,\ 0.5)$$

Threshold $= 0.8$

Predicted labels [1,1,0,0,0,0]
Confusion Matrix

$$TP = 1, \quad FN = 3$$

$$FP = 1, \quad TN = 1$$

Therefore,

$$TPR = \frac{1}{4} = 0.25$$

$$FPR = \frac{1}{2} = 0.5$$

ROC Point

$$(0.5,\ 0.25)$$

Summary of ROC Points

Threshold	TPR	FPR	ROC Point
0.0	1.00	1.00	(1,1)
0.2	1.00	1.00	(1,1)
0.4	0.75	0.50	(0.5,0.75)
0.6	0.50	0.50	(0.5,0.5)
0.8	0.25	0.50	(0.5,0.25)

Each threshold produces one point:

$$(FPR, \ TPR)$$

Joining these points gives the ROC curve.

Summary

- A classifier first predicts **probabilities**, not class labels.
- A **threshold** converts probabilities into binary class labels.
- Changing the threshold changes the **confusion matrix**.
- Each threshold produces one point

$$(FPR,\ TPR)$$

- Joining these points forms the **ROC curve**.
- The ideal classifier is close to the **top-left corner**:

$$(FPR, TPR) = (0, 1)$$

  where
  - High True Positive Rate (TPR)
  - Low False Positive Rate (FPR)
- The diagonal line from $(0, 0)$ to $(1, 1)$ represents a **random classifier**. A good ROC curve should lie above this diagonal.

AUC Calculation Example

Trapezoidal rule for AUC:

$$AUC = \sum \frac{TPR_1 + TPR_2}{2} \times (FPR_2 - FPR_1)$$

Remove the duplicate ROC points

Sort the points and include the starting points(0,0) and endpoints(1,1)

Step 1: Between $(0,0)$ **and** $(0.5, 0.25)$

$$A_1 = (0.5 - 0)\left(\frac{0 + 0.25}{2}\right) = 0.5 \times 0.125 = 0.0625$$

Step 2: Between $(0.5, 0.25)$ **and** $(0.5, 0.50)$

$$A_2 = (0.5 - 0.5)\left(\frac{0.25 + 0.50}{2}\right) = 0$$

Repeat the process for each of the points ‘‘‘

AUC Score

Total Area Under the Curve (AUC)

$$
\begin{aligned}
AUC &= A_1 + A_2 + A_3 + A_4 \\
&= 0.0625 + 0 + 0 + 0.4375 \\
&= \boxed{0.50}
\end{aligned}
$$

AUC = Area Under ROC Curve if between the random range of $0.5 <= AUC <= 1$. Larger AUC better classifier.

AUC	Interpretation
1.0	Perfect
0.9	Excellent
0.8	Good
0.7	Fair
0.5	Random Guess

Higher AUC indicates better classifier.

Regression Performance Metrics

For regression problems:

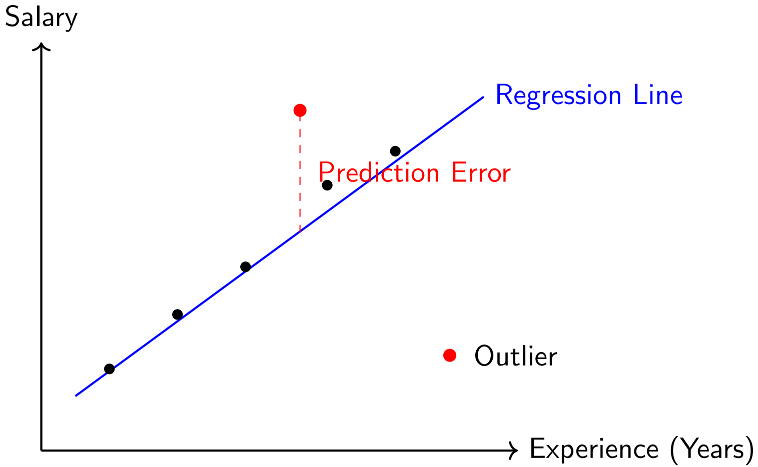
$$y_i = \text{Actual value}$$

$$\hat{y}_i = \text{Predicted value}$$

Common metrics:

1. Mean Squared Error (MSE)
2. Mean Absolute Error (MAE)
3. Root Mean Squared Error (RMSE)
4. R-Squared Score
5. Adjusted R-Squared

Prediction Error in Regression



$$\boxed{\text{Prediction Error (Residual)} = y_i - \hat{y}_i}$$

Prediction Error in Regression

- The regression line predicts the target value.
- The vertical distance between an observed point and the regression line is called the **prediction error (residual)**.
- The objective of regression is to minimize the prediction errors for all data points.

Mean Squared Error (MSE) measures the average squared difference between the actual and predicted values.

$$\boxed{MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where

- n : Number of observations
- y_i : Actual (true) value
- $\hat{y}_i$: Predicted value
- $(y_i - \hat{y}_i)$: Prediction error (Residual)

Advantages and Disadvantages of MSE

Advantages

- Differentiable and easy to optimize using Gradient Descent.
- Convex cost function (for Linear Regression), resulting in a unique global minimum.
- Penalizes large prediction errors, leading to better model fitting.
- Simple to compute and widely used for regression problems.

Disadvantages

- Sensitive to outliers because the errors are squared.
- The error is expressed in squared units (e.g., Rupees^2, Dollars^2), making interpretation difficult.

Note

To express the error in the original unit of the target variable, the **Root Mean Squared Error (RMSE)** is often used:

$$RMSE = \sqrt{MSE}$$

Mean Absolute Error (MAE)

Definition

Mean Absolute Error (MAE) measures the average absolute difference between the actual and predicted values.

$$\boxed{MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|}$$

where

- n : Number of observations
- y_i : Actual (true) value
- $\hat{y}_i$: Predicted value
- $|y_i - \hat{y}_i|$: Absolute prediction error

Advantages and Disadvantages of MAE

Advantages

- **Robust to Outliers**
    - MAE uses the absolute value of the error instead of squaring it.
    - Large errors increase linearly and do not dominate the loss.
- **Easy to Interpret**
    - The error is expressed in the same unit as the target variable.
    - Example: If salary is measured in Rupees, MAE is also measured in Rupees.

Disadvantages

- **Not Differentiable at Zero**
    - The absolute value function has a sharp corner at $x = 0$.
    - Gradient-based optimization becomes more challenging.
- **Optimization May Be Slower**
    - Since MAE is not differentiable at zero, optimization algorithms may require more iterations to converge.

Root Mean Squared Error (RMSE)

Definition

Root Mean Squared Error (RMSE) is the square root of the Mean Squared Error (MSE).

$$\boxed{RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}$$

where

- n : Number of observations
- y_i : Actual (true) value
- $\hat{y}_i$: Predicted value
- $(y_i - \hat{y}_i)^2$: Squared prediction error

Advantages and Disadvantages of RMSE

Advantages

- **Same Unit as the Target Variable**
  - Unlike MSE, RMSE is expressed in the original unit.
  - Example: If salary is measured in Rupees, RMSE is also measured in Rupees.
- **Differentiable**
  - RMSE is smooth and differentiable.
  - It can be optimized efficiently using Gradient Descent.
- **Penalizes Large Errors**
  - Since RMSE is derived from MSE, larger prediction errors receive a higher penalty.

Disadvantages

- **Sensitive to Outliers**
  - Large prediction errors are squared before taking the square root.
  - A few extreme observations can significantly increase the RMSE.

R-Squared (R^2) Score

The **R-Squared (R^2)** score measures how well a regression model explains the variability in the target variable.

$$\boxed{R^2 = 1 - \frac{SS_{res}}{SS_{tot}}}$$

where

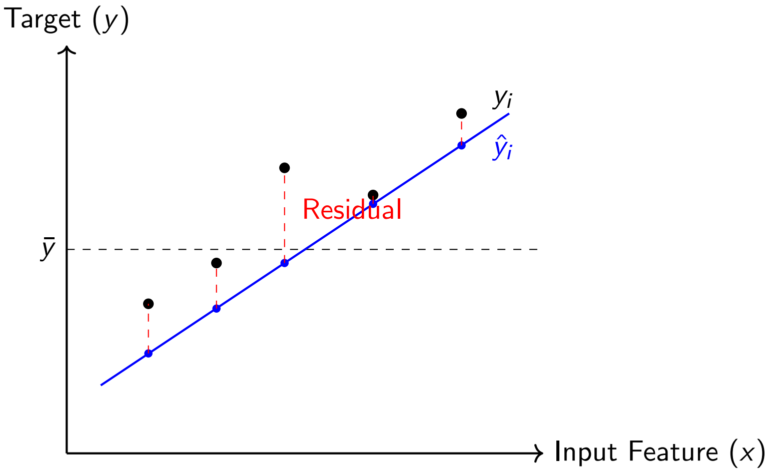
$$SS_{res} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

is the **Residual Sum of Squares (RSS)**, and

$$SS_{tot} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

is the **Total Sum of Squares (TSS)**.

Illustration of R-Squared (R^2)



R-Squared (R^2)

$$R^2 = 1 - \frac{\displaystyle\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\displaystyle\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- $(y_i - \hat{y}_i)^2$: Residual Sum of Squares (Prediction Error)
- $(y_i - \bar{y})^2$: Total Sum of Squares (Variation from the Mean)
- Higher R^2 indicates a better fit of the regression model.

Thank You